# Multiple Forms of Energy in Electromagnetism

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## Energy in Electromagnetism

**All the equations we will discuss in this note:**

$$\begin{cases} W = \dfrac{\epsilon_0}{2}\displaystyle\int_{\mathbb{R}^3} E^2\, d\tau \\ W = \dfrac{1}{2}CV^2 \end{cases} \qquad \begin{cases} W = \dfrac{1}{2\mu_0}\displaystyle\int_{\mathbb{R}^3} B^2\, d\tau \\ W = \dfrac{1}{2}LI^2 \end{cases} \qquad \begin{cases} W = \dfrac{1}{2}\displaystyle\int \mathbf{D}\cdot\mathbf{E}\, d\tau \\ U_{\text{dip}} = -\mathbf{p}\cdot\mathbf{E} \\ U_{\text{dip}} = -\mathbf{m}\cdot\mathbf{B} \end{cases}$$

# 1 The Energy in Electric Fields

## 1.1 Electric Field

The force acting on a charge $q$ in the electric field $\mathbf{E}$ is $q\mathbf{E}$. If we want to "Bring" to the position $\mathbf{r}$ from infinity, we need to always apply the opposite force $-q\mathbf{E}$ to resist the electric force. So, the total work (energy) we do is

$$W = \int \mathbf{F}_{\text{us}}\cdot d\boldsymbol{\ell} = -q\int_{\infty}^{\mathbf{r}} \mathbf{E}\cdot d\boldsymbol{\ell} = -q\int_{\infty}^{\mathbf{r}} \big(-\boldsymbol{\nabla}V\big)\cdot d\boldsymbol{\ell} = qV(\mathbf{r}), \tag{1}$$

where we set $V(\infty) = 0$. Therefore, the work done by us is denoted by:

$$W = qV(\mathbf{r}). \tag{2}$$

As a result, we apply force on the charge and give it energy, then the energy is transferred to the potential energy of the electric field. Now, we want to build a configuration that contains many charges, so we must do works on the charges and transfer the works to be stored in the field.

Suppose we bring the charge $q_i$ from infinity to the position $\mathbf{r}_i$. If there exists charge(s) $q_j$, then we need to do work on $q_i$, that is:

$$W = q_iV(\mathbf{r}_i), \tag{3}$$

where $V$ is the potential[1] due to $q_j$, so we obtain

$$W = q_i \left( \frac{1}{4\pi\epsilon_0} \frac{q_j}{\mathscr{r}_{ij}} \right). \tag{4}$$

The total energy we apply(stored in the field) is the summation all the energy in terms of (4), for every $j$, every $i$:

$$W_{\text{tot}} = \underbrace{\sum_i \sum_{j>i} q_i \left( \frac{1}{4\pi\epsilon_0} \frac{q_j}{\mathscr{r}_{ij}} \right) = \frac{1}{2} \sum_i \sum_{j\neq i} q_i \left( \frac{1}{4\pi\epsilon_0} \frac{q_j}{\mathscr{r}_{ij}} \right)}_{\text{To avoid double counting, we must multiply } 1/2 \text{ or restrict } j>i}. \tag{5}$$

You can look up the slide of the first meeting in Classical Mechanics in the last semester to understand this $\frac{1}{2}$. In general,

$$W = \frac{1}{2} \sum_i q_i \left( \sum_{j\neq i} \frac{1}{4\pi\epsilon_0} \frac{q_j}{\mathscr{r}_{ij}} \right) = \frac{1}{2} \sum_i q_i V(\mathbf{r}_i). \tag{6}$$

When the charges distribute continuously with charge density $\varrho$, we can rewrite (6) in the form:

$$W = \frac{1}{2} \int_{\mathcal{V}} \varrho V \, d\tau = \frac{1}{2} \int_{\mathcal{V}} (\epsilon_0 \boldsymbol{\nabla} \cdot \mathbf{E}) V \, d\tau = \frac{\epsilon_0}{2} \int_{\mathcal{V}} (\boldsymbol{\nabla} \cdot \mathbf{E}) V \, d\tau. \tag{7}$$

By the Products Rules (iv), (7) can be written as

$$\frac{\epsilon_0}{2} \int_{\mathcal{V}} (\boldsymbol{\nabla} \cdot \mathbf{E}) V \, d\tau = \frac{\epsilon_0}{2} \left[ \int_{\mathcal{V}} \boldsymbol{\nabla} \cdot (V\mathbf{E}) \, d\tau - \int_{\mathcal{V}} \mathbf{E} \cdot \underbrace{(\boldsymbol{\nabla} V)}_{=-\mathbf{E}} \, d\tau \right]$$

$$= \frac{\epsilon_0}{2} \oint_{\mathcal{V}} V\mathbf{E} \cdot d\mathbf{a} + \frac{\epsilon_0}{2} \int_{\mathcal{V}} E^2 \, d\tau.$$

Note that the dimensions of $V \sim 1/r$ and $E \sim 1/r^2$, while $da \sim r^2$, which means that $V\mathbf{E} \cdot d\mathbf{a} \sim 1/r$. So, when we take the bounded region $\mathcal{V}$ extremely large ($\mathcal{V} \rightarrow$all space), then the first surface integral will decrease really fast and finally vanish. Hence, the total energy stored in the field is:

$$\boxed{W = \frac{\epsilon_0}{2} \int_{\text{all space}} E^2 \, d\tau.} \tag{8}$$

We can further define the energy density $u_{\text{e}}$

$$u_{\text{e}} = \frac{\epsilon_0}{2} E^2. \tag{9}$$

[1] The form of the potentials (whether electric or magnetic) is mentioned in *Potential and Vector Potential with Multipole Expansion, Yen-Ting Wu*

## 1.2 Capacitance

The capacitance $C$ is defined as

$$C := \frac{Q}{V}, \tag{10}$$

where $V$ is, by definition, the potential of the positive conductor minus that of the negative one. Similarly, the charge $Q$ is on the **positive** conductor (See the appendix of *Polarization & Magnetization, Yen-Ting Wu*). So, the value of the capacitor is always greater than zero, and it represents the ability to store charges in a potential difference $V$. By (2), the energy stored in the capacitor is (you need to do work to transport the next piece of charge):

$$dW = dq \cdot V = dq \left(\frac{q}{C}\right), \tag{11}$$

and integrate the both sides from $q = 0$ to $q = Q$ in a continuous process:

$$\int dW = W = \int_0^Q \frac{q}{C} dq = \frac{1}{2}\frac{Q^2}{C}.$$

Since $Q = CV$ by the definition, we have

$$\boxed{W = \frac{1}{2}CV^2.} \tag{12}$$

This indicates that to charge up a capacitor, we need to "Provide" $W$ to the capacitor with a specific capacitance, and the energy we give will be stored in the capacitor.

# 2 The Energy in Magnetic Fields

## 2.1 Inductance

There are two loops with the constant current, as shown below:

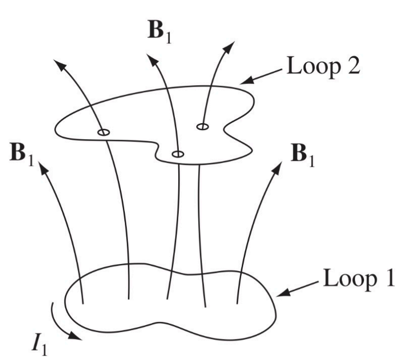


Figure 1: Two loops configuration with the magnetic flux

By Biot-Savart's law, the magnetic field $\mathbf{B}$ of the closed loop is given by

$$\mathbf{B}_1 = \frac{\mu_0 I_1}{4\pi} \oint \frac{d\boldsymbol{\ell}_1 \times \hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}. \tag{13}$$

In addition, the flux through loop 2 is

$$\Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2, \tag{14}$$

and substitute the vector potential $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}_1$ into (14), we have:

$$\Phi_2 = \int \left(\boldsymbol{\nabla} \times \mathbf{A}_1\right) \cdot d\mathbf{a}_2 = \oint \mathbf{A_1} \cdot d\ell_2. \tag{15}$$

The vector potential $\mathbf{A}_1$ is given by the solution to the poisson equation:

$$\mathbf{A}_1 = \frac{\mu_0 I_1}{4\pi} \oint \frac{d\boldsymbol{\ell}_1}{\mathscr{r}},$$

and therefore the flux $\Phi_2$ becomes

$$\Phi_2 = \oint \mathbf{A_1} \cdot d\boldsymbol{\ell}_2 = \left[\frac{\mu_0}{4\pi} \oint \left(\oint \frac{d\boldsymbol{\ell}_1}{\mathscr{r}}\right) \cdot d\boldsymbol{\ell}_2\right] I_1 = \left(\frac{\mu_0}{4\pi} \oint \oint \frac{d\boldsymbol{\ell}_1}{\mathscr{r}} \cdot d\boldsymbol{\ell}_2\right) I_1. \tag{16}$$

It implies that the flux is proportional to the amplitude of the current, no matter what shape the loop takes. So we can say that

$$\Phi = \alpha I, \tag{17}$$

where $\alpha$ is the proportionality. In this case, $\Phi_2 = \alpha_{21} I_1$, where

$$\alpha_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\boldsymbol{\ell}_1}{\mathscr{r}} \cdot d\boldsymbol{\ell}_2 = \frac{\mu_0}{4\pi} \oint \oint \frac{d\boldsymbol{\ell}_2}{\mathscr{r}} \cdot d\boldsymbol{\ell}_1 = \alpha_{12}. \tag{18}$$

In conclusion, under the same configuration, or with the specific current and magnetic field, the proportionalities are the same. This leads to the emf($\mathcal{E}$) when one of the currents varies in time with the same constant. By flux law,

$$\mathcal{E}_2 = -\frac{d\Phi_2}{dt} = -\alpha \frac{dI_1}{dt}. \tag{19}$$

It also means that the changing current $dI_1$ will induce the current (or emf) of loop 2. However, the loop can also induce an emf by itself; the flux through its own loop is also proportional to its current. Here, we define the proportionality as $L$, which is called **Inductance**, it satisfies

$$\Phi = LI. \tag{20}$$

When the current changes, the emf induced in the loop itself is:

$$\mathcal{E} = -\frac{d\Phi}{dt} = -L\frac{dI}{dt}. \tag{21}$$

Now, we can finally start to discuss the energy of the inductance and the energy stored in the magnetic field. The work $dW$ done by us in an infinitesimal time on the charge through the loop is:

$$dW = \underbrace{-\oint d\mathbf{F}\cdot d\boldsymbol{\ell}}_{\substack{\text{Minus sign means that the}\\ \text{work done by us is}\\ \text{against the emf}}} = -dq\oint \mathbf{E}\cdot d\ell$$

$$\Rightarrow \frac{dW}{dt} = -\underbrace{I}_{dq/dt}\oint \mathbf{E}\cdot d\boldsymbol{\ell} = -\mathcal{E}I,$$

so the work done per unit time is:

$$\frac{dW}{dt} = LI\frac{dI}{dt} \Leftrightarrow dW = LI\,dI. \tag{22}$$

During the process of doing work, the current gradually increases. As a result, the resistance we must overcome becomes larger, and therefore, more energy is required. We start with $I = 0$ and build it up to $I$, then the total work done by us $W_{\text{us}}$, which is also stored in the field is:

$$\boxed{W_{\mathbf{us}} = \int_0^I LI'\,dI' = \frac{1}{2}LI^2.} \tag{23}$$

## 2.2 Magnetic Field

We've known that the flux through the loop itself is

$$\Phi = \int \underbrace{\mathbf{B}}_{\text{Caused by loop}} \cdot d\mathbf{a} = \oint \mathbf{A}\cdot d\boldsymbol{\ell} = LI$$

Combining this with (23), we have

$$W = \frac{1}{2}I\oint \mathbf{A}\cdot d\boldsymbol{\ell} = \frac{1}{2}\oint \mathbf{A}\cdot\mathbf{I}\,d\ell. \tag{24}$$

In general,

$$W = \frac{1}{2}\int_{\mathcal{V}} \mathbf{A}\cdot\mathbf{J}\,d\tau = \frac{1}{2}\int_{\mathcal{V}} \underbrace{\mathbf{A}\cdot\left(\frac{1}{\mu_0}\boldsymbol{\nabla}\times\mathbf{B}\right)}_{\text{Faraday's Law without changing }\mathbf{E}} d\tau = \frac{1}{2\mu_0}\int_{\mathcal{V}} \mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B})d\tau. \tag{25}$$

Apply the Products Rules (iii), (25) can be written as

$$\begin{aligned}\frac{1}{2\mu_0}\int_{\mathcal{V}} \mathbf{A}\cdot(\boldsymbol{\nabla}\times\mathbf{B})d\tau &= \frac{1}{2\mu_0}\left[\int_{\mathcal{V}} \mathbf{B}\cdot\Big(\underbrace{\boldsymbol{\nabla}\times\mathbf{A}}_{=\mathbf{B}}\Big)d\tau - \int_{\mathcal{V}} \boldsymbol{\nabla}\cdot(\mathbf{A}\times\mathbf{B})d\tau\right] \\ &= \frac{1}{2\mu_0}\left[\int_{\mathcal{V}} B^2\,d\tau - \oint_{\partial\mathcal{V}} (\mathbf{A}\times\mathbf{B})\cdot d\mathbf{a}\right].\end{aligned}$$

Likewise, we note that the dimensions of $A \sim 1/r$ and $B \sim 1/r^2$, while $da \sim r^2$, which means that $(\mathbf{A}\times\mathbf{B})\cdot d\mathbf{a} \sim 1/r$. So, when we take the bounded region $\mathcal{V}$ extremely large ($\mathcal{V} \to$all space), then the second surface integral will decrease really fast and finally vanish. Hence, the total energy stored in the field is:

$$\boxed{W = \frac{1}{2\mu_0}\int_{\text{all space}} B^2 d\tau.} \tag{26}$$

We can further define the energy density $u_{\mathrm{m}}$ in the magnetic field:

$$u_{\mathrm{m}} = \frac{1}{2\mu_0}B^2 \tag{27}$$

# 3 The Energy of a Dipole and a Linear Dielectric in Ideal Dipole Systems

## 3.1 Electric Dipole

The charge $-q$ is located at position $\mathbf{r}$ relative to the origin, while the charge $+q$ is located at $\mathbf{r}'$. The vector $\mathbf{d}$ denotes the displacement from $-q$ to $+q$, such that $\mathbf{r}' = \mathbf{r} + \mathbf{d}$. As shown below

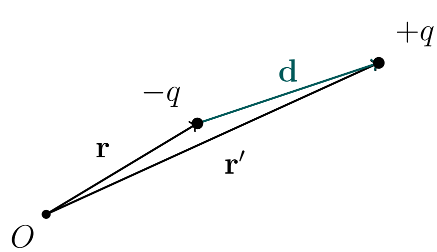


Figure 2: Geometry of an electric dipole.

So, the total energy stored in this system is:

$$
\begin{aligned}
U &= qV(\mathbf{r}+\mathbf{d}) + \Big[-qV(\mathbf{r})\Big] \\
&= q\Big[V(\mathbf{r}+\mathbf{d}) - V(\mathbf{r})\Big] \\
&= q\left[-\int_{\mathbf{r}}^{\mathbf{r}+\mathbf{d}} \mathbf{E}\cdot d\boldsymbol{\ell}\right] = q(-\mathbf{E}\cdot\mathbf{d}) = -\mathbf{E}\cdot q\mathbf{d},
\end{aligned}
$$

for an ideal dipole. Remember that $\mathbf{p} = q\mathbf{d}$ So, the energy $U$ can be written as:

$$
\boxed{U = -\mathbf{p}\cdot\mathbf{E}.} \tag{28}
$$

## 3.2 Magnetic Dipole

Consider a small current loop carrying steady current $I$. The magnetic force acting on a line element $d\boldsymbol{\ell}$ is

$$
d\mathbf{F} = I\, d\boldsymbol{\ell} \times \mathbf{B}(\mathbf{r}).
$$

Hence the total force on the loop is

$$
\mathbf{F} = I \oint d\boldsymbol{\ell} \times \mathbf{B}(\mathbf{r}). \tag{29}
$$

To extract the dipole approximation, choose a reference point $\mathbf{r}_0$ inside the loop and write

$$
\mathbf{r} = \mathbf{r}_0 + \boldsymbol{\alpha},
$$

where $\boldsymbol{\alpha}$ denotes the displacement from the reference point to the current element. Since the loop is assumed to be small, expand the magnetic field around $\mathbf{r}_0$:

$$
\mathbf{B}(\mathbf{r}) = \mathbf{B}(\mathbf{r}_0) + (\boldsymbol{\alpha}\cdot\boldsymbol{\nabla})\mathbf{B}(\mathbf{r}_0) + \cdots . \tag{30}
$$

Substituting this into the force integral gives

$$
\mathbf{F} = I \oint d\boldsymbol{\ell} \times \left[\mathbf{B}(\mathbf{r}_0) + (\boldsymbol{\alpha}\cdot\boldsymbol{\nabla})\mathbf{B}(\mathbf{r}_0)\right].
$$

The first term vanishes because $\mathbf{B}(\mathbf{r}_0)$ is constant over the loop:

$$
I \oint d\boldsymbol{\ell} \times \mathbf{B}(\mathbf{r}_0) = I\left(\oint d\boldsymbol{\ell}\right) \times \mathbf{B}(\mathbf{r}_0) = 0.
$$

Therefore, to first order,

$$
\mathbf{F} = I \oint d\boldsymbol{\ell} \times \left[(\boldsymbol{\alpha}\cdot\boldsymbol{\nabla})\mathbf{B}\right].
$$

Now write this in index notation:

$$F_i = I \oint \epsilon_{ijk}\, d\ell_j\, \alpha_l\, \partial_l B_k. \tag{31}$$

Since $\partial_l B_k$ is evaluated at $\mathbf{r}_0$, it is independent of the integration variable and can be taken outside the integral:

$$F_i = I\, \epsilon_{ijk}\, \partial_l B_k \oint \alpha_l\, d\ell_j.$$

Next, only the antisymmetric part of $\alpha_l d\ell_j$ contributes, because it is contracted with $\epsilon_{ijk}$. Thus,

$$\oint \alpha_l\, d\ell_j \;\longrightarrow\; \frac{1}{2}\oint \left(\alpha_l\, d\ell_j - \alpha_j\, d\ell_l\right).$$

Hence,

$$F_i = \frac{I}{2}\epsilon_{ijk}\partial_l B_k \oint \left(\alpha_l\, d\ell_j - \alpha_j\, d\ell_l\right).$$

Now introduce the area vector of the loop:

$$a_m = \frac{1}{2}\oint (\boldsymbol{\alpha} \times d\boldsymbol{\ell})_m = \frac{1}{2}\epsilon_{mjl}\oint \alpha_j\, d\ell_l.$$

Equivalently,

$$\oint \left(\alpha_l\, d\ell_j - \alpha_j\, d\ell_l\right) = 2\, \epsilon_{ljm}\, a_m.$$

Substituting this into the force expression,

$$F_i = \frac{I}{2}\epsilon_{ijk}\partial_l B_k \left(2\epsilon_{ljm} a_m\right),$$

so that

$$\begin{aligned} F_i &= I\, \epsilon_{ijk}\epsilon_{ljm}\, \partial_l B_k\, a_m \\ &= I(\delta_{il}\delta_{km} - \delta_{im}\delta_{kl})\partial_l B_k\, a_m \\ &= I\, a_m\, \partial_i B_m - I\, a_i\, \partial_k B_k. \end{aligned}$$

By Maxwell's equation

$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0,$$

the second term vanishes, and therefore

$$F_i = I\, a_m\, \partial_i B_m. \tag{32}$$

Finally, define the magnetic dipole moment

$$\mathbf{m} = I\, \mathbf{a}, \qquad m_m = I a_m.$$

Then

$$F_i = m_m \, \partial_i B_m. \tag{33}$$

Since $\mathbf{m}$ is treated as constant with respect to spatial differentiation,

$$\partial_i(\mathbf{m} \cdot \mathbf{B}) = \partial_i(m_m B_m) = m_m \, \partial_i B_m.$$

Hence we conclude

$$\mathbf{F} = \boldsymbol{\nabla}(\mathbf{m} \cdot \mathbf{B}). \tag{34}$$

So, the energy stored in the magnetic dipole field is:

$$U = -\int_{\infty}^{\mathbf{r}} \mathbf{F} \cdot d\boldsymbol{\ell} = -\int_{\infty}^{\mathbf{r}} \boldsymbol{\nabla}(\mathbf{m} \cdot \mathbf{B}) \cdot d\boldsymbol{\ell}, \tag{35}$$

where the magnetic field at $\mathbf{r} = \infty$ is zero, so:

$$\boxed{U = -\mathbf{m} \cdot \mathbf{B}.} \tag{36}$$

## 3.3 Linear Medium

By (7), we can obtain the infinitesimal difference of the energy $dW$ in a linear medium:

$$dW = \int (d\varrho_f) V \, d\tau = \int \boldsymbol{\nabla} \cdot (d\mathbf{D}) V \, d\tau.$$

This can be denoted by the index notation:

$$\begin{aligned} dW &= \int V \partial_i (dD)_i \, d\tau \quad V \partial_i (dD)_i = \partial_i (V dD_i) - (\partial_i V)(dD)_i \\ &= \int \partial_i (V dD_i) d\tau - \int (\underbrace{\partial_i V}_{=-E_i})(dD)_i d\tau \\ &= \oint V dD_i \, d\tau + \int E_i (dD)_i \, d\tau. \end{aligned}$$

The boundary term will vanish because of the dimensions (recall the discussion in the energy of the electric & magnetic fields). Because its a linear dielectric, the term $(dD)_i = (d\epsilon E)_i = \epsilon (dE)_i$. Therefore, we have

$$E_i dD_i = \epsilon (E_i dE_i) = \frac{1}{2} \epsilon d(E_i E_i) = d\left(\frac{1}{2} E_i D_i\right). \tag{37}$$

Substituting (37) into $dW$, we can obtain

$$dW = \int d\left(\frac{1}{2} E_i D_i\right) d\tau \Rightarrow W = \frac{1}{2} \int E_i D_i \, d\tau, \tag{38}$$

as a result, the energy can be written as

$$\boxed{W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} \, d\tau.} \tag{39}$$

## Some Identities

**Triple Products:**

$$\begin{cases} \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \\ \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad \text{Bac(k)-Cab law} \end{cases}$$

**Products Rules:**

$$\begin{cases} \boldsymbol{\nabla}(fg) = f(\boldsymbol{\nabla} g) + g(\boldsymbol{\nabla} f) \\ \boldsymbol{\nabla}(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\boldsymbol{\nabla} \times \mathbf{B}) + \mathbf{B}(\boldsymbol{\nabla} \times \mathbf{A}) + (\mathbf{A} \cdot \boldsymbol{\nabla})\mathbf{B} + (\mathbf{B} \cdot \boldsymbol{\nabla})\mathbf{A} \\ \boldsymbol{\nabla} \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) - \mathbf{A} \cdot (\boldsymbol{\nabla} \times \mathbf{B}) \\ \boldsymbol{\nabla} \cdot (f\mathbf{A}) = f(\boldsymbol{\nabla} \cdot \mathbf{A}) + \mathbf{A} \cdot \boldsymbol{\nabla} f \\ \boldsymbol{\nabla} \times (f\mathbf{A}) = f(\boldsymbol{\nabla} \times \mathbf{A}) - \mathbf{A} \times \boldsymbol{\nabla} f \\ \boldsymbol{\nabla} \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \boldsymbol{\nabla})\mathbf{A} - (\mathbf{A} \cdot \boldsymbol{\nabla})\mathbf{B} + \mathbf{A}(\boldsymbol{\nabla} \cdot \mathbf{B}) - \mathbf{B}(\boldsymbol{\nabla} \cdot \mathbf{A}) \end{cases}$$

**Second Derivatives:**

$$\begin{cases} \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) = 0 \\ \boldsymbol{\nabla} \times (\boldsymbol{\nabla} f) = \mathbf{0} \\ \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \end{cases}$$

**Green's Identities and Some Integrals:**

$$\begin{cases} \displaystyle\int_{\mathcal{V}} \left(T\,\nabla^2 U - U\nabla^2 T\right) d\tau = \oint_{\mathcal{S}} \left(T\boldsymbol{\nabla} U - U\boldsymbol{\nabla} T\right)\cdot d\mathbf{a} \\ \displaystyle\int_{\mathcal{S}} \boldsymbol{\nabla} T \times d\mathbf{a} = -\oint_{\mathcal{P}} T\,d\boldsymbol{\ell} \\ \displaystyle\oint (\mathbf{c}\cdot\mathbf{r})\,d\boldsymbol{\ell} = \left(\int d\mathbf{a}\right)\times\mathbf{c} \\ \displaystyle\int_{\mathcal{V}} \boldsymbol{\nabla}\times\mathbf{F}\,d\tau = -\oint_{\mathcal{S}} \mathbf{F}\times d\mathbf{a} \end{cases}$$

**Delta functions:**

$$\begin{cases} \displaystyle\boldsymbol{\nabla}\cdot\left(\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}\right) = 4\pi\delta^3(\boldsymbol{\mathscr{r}}) \\ \displaystyle\boldsymbol{\nabla}\left(\frac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla}\left(\frac{1}{\mathbf{r}-\mathbf{r}'}\right) = -\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\ \displaystyle\boldsymbol{\nabla}'\left(\frac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla}'\left(\frac{1}{\mathbf{r}-\mathbf{r}'}\right) = \frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\ \displaystyle\nabla^2\left(\frac{1}{\mathscr{r}}\right) = -4\pi\delta^3(\boldsymbol{\mathscr{r}}) \end{cases}$$